Design IC Final

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Target: systolic array – matrix multiplier (4x4 matrix with 8 32-bit inputs, 16 outputs)

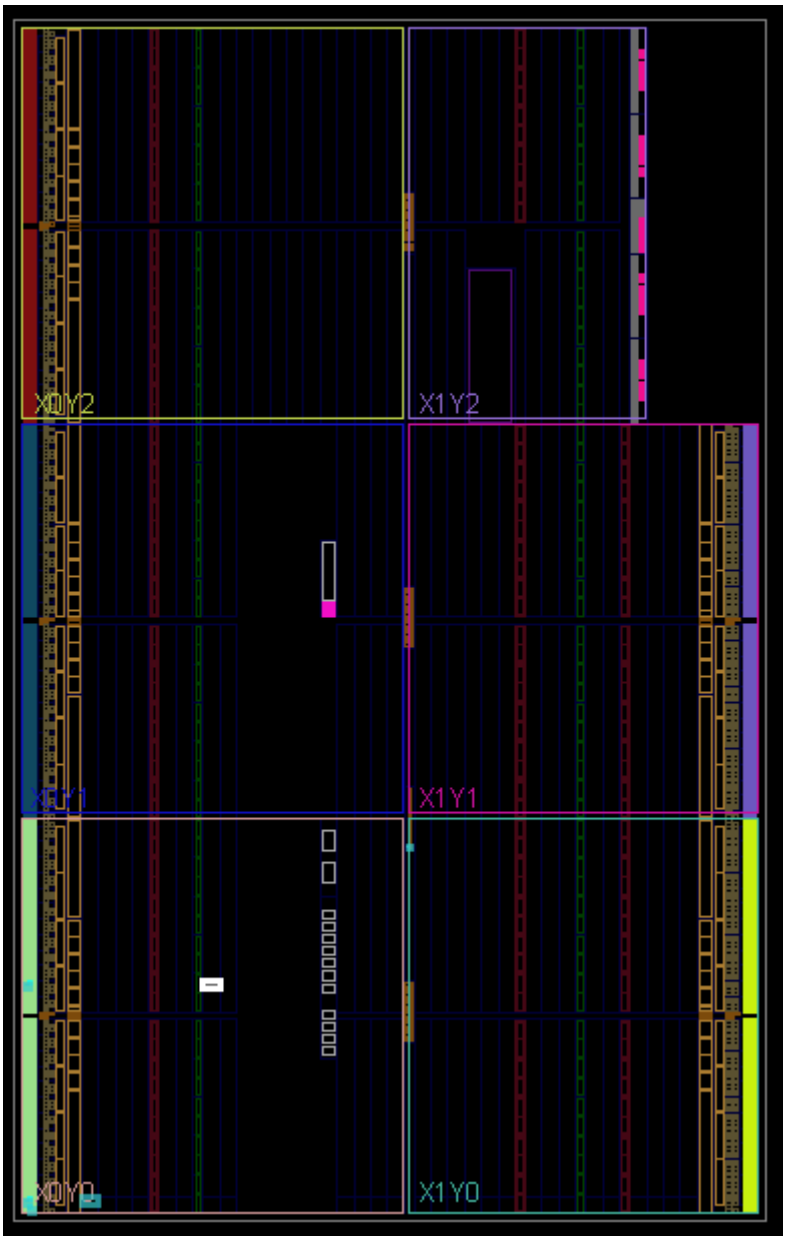
On process

- Process: ASAP7
- Power: 584.5 [uW]
- Area: 123919.735737 [um^2]
- Frequency: 500 [MHz]
- Critical path: inp_west4[1] -> P4/result_reg[63]
- Slack: 0.02 [ns] (MET)

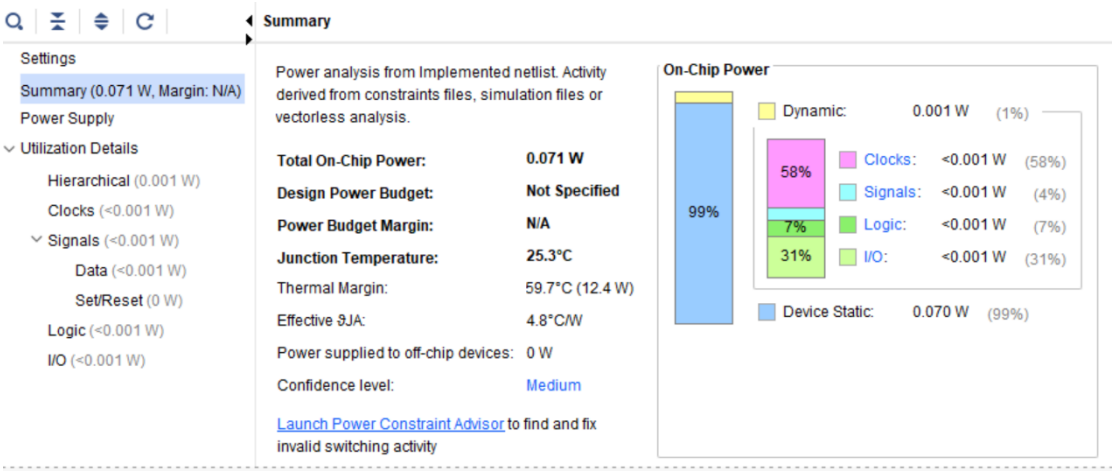
On board

- Tool: Xilinx/Vivado
- Board:xc7a35tcsg324-1
- Power:71 [mW]
- Frequency: 100 [MHz]
- Slack: 0.29 [ns] (MET)

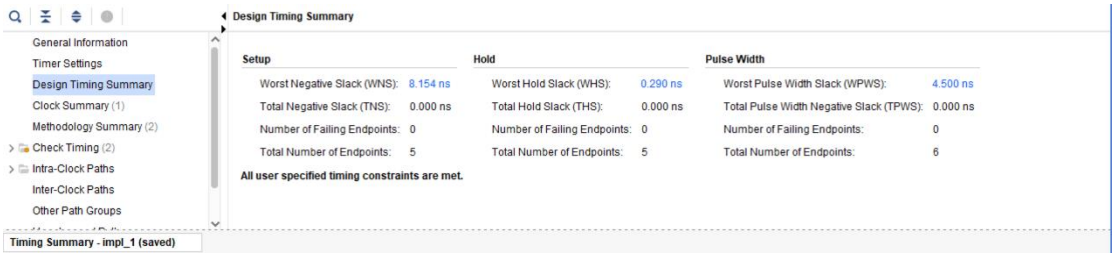
Layout – on board



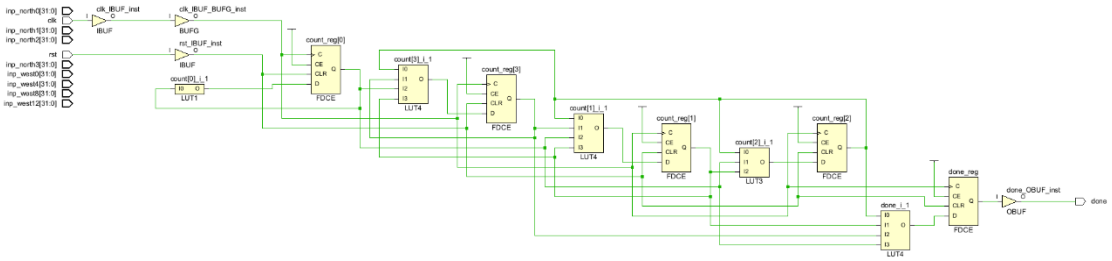
Power – on board



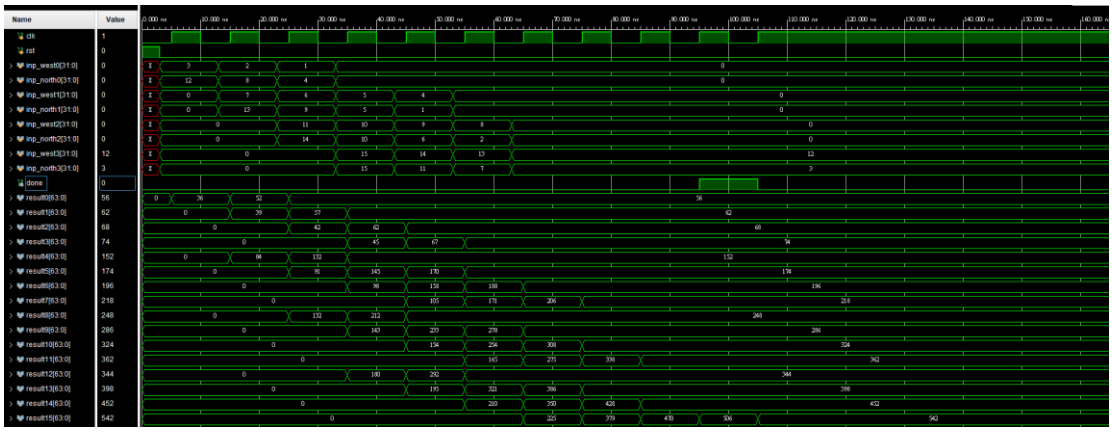
Slack – on board



Schematic – on board



Wave - On process



Timing - On process

```
clock clk (rise edge)                                   2000.00    2000.00
clock network delay (ideal)                                0.00    2000.00
P4/result_reg[63]/CLK (ASYNC_DFFHx1_ASAP7_75t_R)           0.00    2000.00 r
library setup time                                       -16.60    1983.40
data required time                                                 1983.40
-------------------------------------------------------------------------
data required time                                                 1983.40
data arrival time                                                 -1983.39
-------------------------------------------------------------------------
slack (MET)                                                           0.02
```

Area - On process

```

****************************************
Report : area
Design : systolic_array
Version: T-2022.03
Date   : Tue Dec 19 15:29:08 2023
****************************************

Library(s) Used:

    asap7sc7p5t_INVBUF_RVT_TT_08302018 (File: /RAID2/COURSE/dic/dic112/dic/NLDM/asap7sc7p5t_INVBUF_RVT_TT_08302018.db)
    asap7sc7p5t_SIMPLE_RVT_TT_08302018 (File: /RAID2/COURSE/dic/dic112/dic/NLDM/asap7sc7p5t_SIMPLE_RVT_TT_08302018.db)
    asap7sc7p5t_SEQ_RVT_TT_08302018 (File: /RAID2/COURSE/dic/dic112/dic/NLDM/asap7sc7p5t_SEQ_RVT_TT_08302018.db)

Number of ports:                          8515
Number of nets:                          96910
Number of cells:                         83429
Number of combinational cells:           81328
Number of sequential cells:               2053
Number of macros/black boxes:                0
Number of buf/inv:                       12545
Number of references:                       25

Combinational area:              111467.715978
Buf/Inv area:                      9421.479478
Noncombinational area:            12452.019759
Macro/Black Box area:                 0.000000
Net Interconnect area:      undefined  (No wire load specified)

Total cell area:                 123919.735737
Total area:                 undefined
1

```

Power - On process

```
****************************************
Report : Time Based **Power**
Design : systolic_array
Version: P-2019.03-SP5-1
Date   : Tue Dec 19 15:33:42 2023
****************************************



  Attributes
  ----------
      i  -  Including register clock pin internal power
      u  -  User defined power group

                         Internal  Switching  Leakage    Total
Power Group              Power     Power      Power      Power   (     %)  Attrs
--------------------------------------------------------------------------------
clock_network               0.0000    0.0000    0.0000    0.0000 ( 0.00%)
register                 2.620e-04 1.628e-06 5.928e-07 2.642e-04 (45.21%)  i
combinational            1.373e-04 1.752e-04 7.675e-06 3.203e-04 (54.79%)
sequential                  0.0000    0.0000    0.0000    0.0000 ( 0.00%)
memory                      0.0000    0.0000    0.0000    0.0000 ( 0.00%)
io_pad                      0.0000    0.0000    0.0000    0.0000 ( 0.00%)
black_box                   0.0000    0.0000    0.0000    0.0000 ( 0.00%)

  Net Switching Power  = 1.769e-04   (30.26%)
  Cell Internal Power  = 3.993e-04   (68.33%)
  Cell Leakage Power   = 8.268e-06   ( 1.41%)
    Intrinsic Leakage  = 8.268e-06
    Gate Leakage       =    0.0000
                         ---------
Total Power            = 5.845e-04  (100.00%)

X Transition Power     =    0.0000
Glitching Power        = 4.811e-06

Peak Power             = 7.274e-03
Peak Time              =        45
```

Code (verilog)

Block.v

```
module block(inp_north, inp_west, clk, rst, outp_south, outp_east, result);
    input [31:0] inp_north, inp_west;
    output reg [31:0] outp_south, outp_east;
    input clk, rst;
    output reg [63:0] result;
    wire [63:0] multi;
    always @(posedge rst or posedge clk) begin
        if(rst) begin
            result <= 0;
            outp_east <= 0;
            outp_south <= 0;
        end
        else begin
            result <= result + multi;
            outp_east <= inp_west;
            outp_south <= inp_north;
        end
    end
    assign multi = inp_north*inp_west;
endmodule
```

systolic_array.v

```
`include "block.v"
module systolic_array(inp_west0, inp_west4, inp_west8, inp_west12,
              inp_north0, inp_north1, inp_north2, inp_north3,
              clk, rst, done);

    input [31:0] inp_west0, inp_west4, inp_west8, inp_west12,
              inp_north0, inp_north1, inp_north2, inp_north3;
    output reg done;
    input clk, rst;
    reg [3:0] count;



    wire [31:0] inp_north0, inp_north1, inp_north2, inp_north3;
    wire [31:0] inp_west0, inp_west4, inp_west8, inp_west12;
    wire [31:0] outp_south0, outp_south1, outp_south2, outp_south3,
outp_south4, outp_south5, outp_south6, outp_south7, outp_south8,
outp_south9, outp_south10, outp_south11, outp_south12, outp_south13,
outp_south14, outp_south15;
    wire [31:0] outp_east0, outp_east1, outp_east2, outp_east3, outp_east4,
outp_east5, outp_east6, outp_east7, outp_east8, outp_east9, outp_east10,
outp_east11, outp_east12, outp_east13, outp_east14, outp_east15;
    wire [63:0] result0, result1, result2, result3, result4, result5,
result6, result7, result8, result9, result10, result11, result12, result13,
result14, result15;



    //from north and west
    block P0 (inp_north0, inp_west0, clk, rst, outp_south0, outp_east0,
result0);
    //from north
    block P1 (inp_north1, outp_east0, clk, rst, outp_south1, outp_east1,
result1);
    block P2 (inp_north2, outp_east1, clk, rst, outp_south2, outp_east2,
result2);
    block P3 (inp_north3, outp_east2, clk, rst, outp_south3, outp_east3,
```

```
result3);

    //from west
    block P4 (outp_south0, inp_west4, clk, rst, outp_south4, outp_east4,
result4);
    block P8 (outp_south4, inp_west8, clk, rst, outp_south8, outp_east8,
result8);
    block P12 (outp_south8, inp_west12, clk, rst, outp_south12,
outp_east12, result12);

    //no direct inputs
    //second row
    block P5 (outp_south1, outp_east4, clk, rst, outp_south5, outp_east5,
result5);
    block P6 (outp_south2, outp_east5, clk, rst, outp_south6, outp_east6,
result6);
    block P7 (outp_south3, outp_east6, clk, rst, outp_south7, outp_east7,
result7);
    //third row
    block P9 (outp_south5, outp_east8, clk, rst, outp_south9, outp_east9,
result9);
    block P10 (outp_south6, outp_east9, clk, rst, outp_south10,
outp_east10, result10);
    block P11 (outp_south7, outp_east10, clk, rst, outp_south11,
outp_east11, result11);
    //fourth row
    block P13 (outp_south9, outp_east12, clk, rst, outp_south13,
outp_east13, result13);
    block P14 (outp_south10, outp_east13, clk, rst, outp_south14,
outp_east14, result14);
    block P15 (outp_south11, outp_east14, clk, rst, outp_south15,
outp_east15, result15);

    always @(posedge clk or posedge rst) begin
        if(rst) begin
            done <= 0;
            count <= 0;
        end
```

```
        else begin
            if(count == 9) begin
                done <= 1;
                count <= 0;
            end
            else begin
                done <= 0;
                count <= count + 1;
            end
        end
    end

endmodule
```

systolic_array_tb.v

```
`include "systolic_array.v"
module sys_array_tb;

reg rst, clk;

reg [31:0] inp_west0, inp_west1, inp_west2, inp_west3, inp_north0,
inp_north1, inp_north2, inp_north3;
wire done;

systolic_array uut(inp_west0, inp_west1, inp_west2, inp_west3,
               inp_north0, inp_north1, inp_north2, inp_north3,
               clk, rst, done);


initial begin
    #3  inp_west0 <= 32'd3;
        inp_north0 <= 32'd12;
    #10 inp_west0 <= 32'd2;
        inp_north0 <= 32'd8;
    #10 inp_west0 <= 32'd1;
        inp_north0 <= 32'd4;
    #10 inp_west0 <= 32'd0;
        inp_north0 <= 32'd0;
    #10 inp_west0 <= 32'd0;
        inp_north0 <= 32'd0;
    #10 inp_west0 <= 32'd0;
        inp_north0 <= 32'd0;
    #10 inp_west0 <= 32'd0;
        inp_north0 <= 32'd0;
end

initial begin
    #3  inp_west1 <= 32'd0;
        inp_north1 <= 32'd0;
    #10 inp_west1 <= 32'd7;
        inp_north1 <= 32'd13;
    #10 inp_west1 <= 32'd6;
```

```
        inp_north1 <= 32'd9;
    #10 inp_west1 <= 32'd5;
        inp_north1 <= 32'd5;
    #10 inp_west1 <= 32'd4;
        inp_north1 <= 32'd1;
    #10 inp_west1 <= 32'd0;
        inp_north1 <= 32'd0;
    #10 inp_west1 <= 32'd0;
        inp_north1 <= 32'd0;
end

initial begin
    #3  inp_west2 <= 32'd0;
        inp_north2 <= 32'd0;
    #10 inp_west2 <= 32'd0;
        inp_north2 <= 32'd0;
    #10 inp_west2 <= 32'd11;
        inp_north2 <= 32'd14;
    #10 inp_west2 <= 32'd10;
        inp_north2 <= 32'd10;
    #10 inp_west2 <= 32'd9;
        inp_north2 <= 32'd6;
    #10 inp_west2 <= 32'd8;
        inp_north2 <= 32'd2;
    #10 inp_west2 <= 32'd0;
        inp_north2 <= 32'd0;
end

initial begin
    #3  inp_west3 <= 32'd0;
        inp_north3 <= 32'd0;
    #10 inp_west3 <= 32'd0;
        inp_north3 <= 32'd0;
    #10 inp_west3 <= 32'd0;
        inp_north3 <= 32'd0;
    #10 inp_west3 <= 32'd15;
        inp_north3 <= 32'd15;
    #10 inp_west3 <= 32'd14;
```

```
        inp_north3 <= 32'd11;
    #10 inp_west3 <= 32'd13;
        inp_north3 <= 32'd7;
    #10 inp_west3 <= 32'd12;
        inp_north3 <= 32'd3;
end

initial begin
rst <= 1;
clk <= 0;
#3
rst <= 0;
end

initial begin
    repeat(21)
         #5 clk <= ~clk;
end




initial begin
    $dumpfile("wave.vcd");
    $dumpvars(0, sys_array_tb);
end



endmodule
```